

Database Ideas

- Integrate directly with UNT system directly
- Copy from UNT database

Users

- Let users individually select schedules or take directly from UNT systems

Pathing

- 2D pathing within buildings
 - Hand-trace or import the outdoor path network as GeoJSON, with building entrance nodes
 - Load into NetworkX, run Dijkstra with distance weights, verify paths look sane
 - Render with Leaflet or SVG, static weights only
 - Convert weights to seconds, add a time parameter to the query
 - Layer in schedule-driven crowding multipliers
 - Add indoor penalties, then real indoor graphs for one building
 - Add route alternatives and accessibility mode

Listed features

Models things like:

- Buildings
- Classrooms
- energy usage
- Parking
- Traffic
- Occupancy
- Events
- Maintenance
- environmental conditions

Project Skeleton

```
campus-twin/
├─ backend/
│  └─ app/
│     │ └─ main.py           # FastAPI app, CORS, router mounting
│     │ └─ api/routes/       # /route, /buildings, /schedule
│     │ └─ core/             # graph loading, Dijkstra, weighting
│     │ └─ models/           # Pydantic request/response schemas
│     │ └─ data/             # GeoJSON: paths, buildings, entrances
│     └─ tests/
├─ frontend/                 # Vite + React + TypeScript
│  └─ src/
│     │ └─ components/Map/
│     │ └─ api/              # generated or hand-written client
│     └─ types/
└─ docker-compose.yml        # optional, add when you need it
```